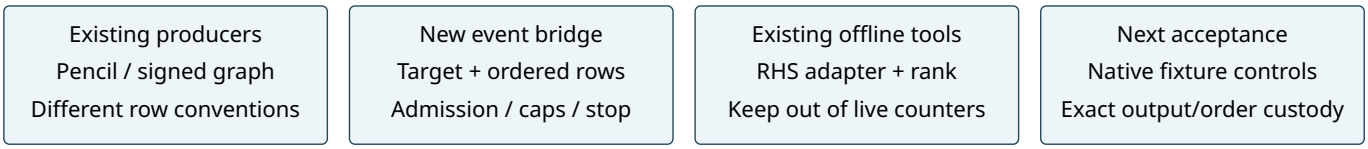
Native target-local relation stream: next construction

Source inspection at published revision 5342909fbcaffae8f8bce2b290bdc095d47eccb6 identifies existing rank, sign-conversion and duplicate-label components and the missing production event bridge. Fifteen exact source records are bound in source-bindings.json. This is a source-only reconciliation: no builds, scientific data, cloud actions or frozen-input changes. No other reviewers' reports were read in this task.



Already implemented. rank-information/src/lib.rs:38 computes offline consistency and target identifiability using three rank calculations; relation-adapter/src/lib.rs:6 already converts augmented RHS rows. The native row-sign checker exists but accepts one nuisance column and fixed stdin rows; its retained producer status records zero actual collector streams. TargetInformation.unique_iff_outside_range is formal source for the consistent affine-system criterion. SignedGraph::target_rank_score and its rank-policy callsite already exist; they must not be proposed as missing. The duplicate Labels::ingest table already sign-normalizes both labels and verifies a candidate by kG=Q, but its inspected callers are planted fixtures and retained archival cases.

Concrete bridge point. The ecbench pencil adapter includes research/pencil_large_prime_20261029/src/native_solver.rs. Its accepted geometric relation enters GraphSolver::relation at lines1357-1359; sparse terms are assembled at471-492 and passed to Collector::insert. Capture that exact shared assembly result and its Admission before/after insertion, along with target identity and actual walk_index/lambda order. Collector cap outcomes retain pending constraints; distinguish generated, incorporated and residual-emitted rows. A later signed-graph adapter must also capture earlier duplicate/identity bypasses and each considered candidate, not only chosen walk states.

Two conventions. Pencil stores augmented [A,b,-a], meaning Ax+bk=-a; its zero-equation row is [A,b,+a]. SignedGraph stores su*xu+sv*xv=a+bk; its zero-equation row is [A,-b,-a], with known-zero node0 excluded from nuisance variables. A typed row-convention field and controls must prevent mixing them. Preserve dynamic column maps and zero extension of older prefixes. Actual scalar verification stays with the native group check.

Implementation obligations. Add an optional bounded native event observer with immutable target/attempt header, monotonic event ordinal, source/workload/curve/order bindings, eligibility and duplicate dispositions, exact sparse row, Admission/cap/terminal result and first-label references. Keep the no-observer algorithm path and stop order. Add an offline Rust loader/replayer using existing sign and rank components. It must reject target changes, reordered/truncated events, wrong signs, stale columns, invalid domains and cap-pending rows presented as incorporated. Record serialization, storage and observer failure explicitly.

Accounting boundary. The existing information() rebuilds three bases and changes shared thread-local counters: it belongs outside the measured producer and must not be inserted online or paired with mid-run counter resets. Rank score and global target-identifiability are distinct objects to cross-check. Natural duplicate/collision streams remain absent from the inspected inventory; archived420 cases and planted alternate labels cannot supply a natural rate. New timing or cost comparison requires its own qualified protocol and execution limits.

Source map and next edit. SOURCE-MAP.md supplies exact paths, entrypoints, contracts and acceptance controls. source-bindings.json records each source SHA256 and last-change commit at the pinned snapshot; construction.json distinguishes existing code from proposed hooks. A scoped next task can implement pencil sparse-row events plus offline replay first, while retaining signed/Kummer hooks as explicit subsequent obligations. The graph is a construction map, not a performance plot.